Cheatsheets / **Learn Lua**

codecademy

Conditionals & Logic

Else Statements

Lua `else` statements will execute a code block if every expression in the if-based control structure evaluates to `false`. If an `else` is included in a control structure, the `else` must go at the end.

```
if isSunny then
  print("Go outside!")
else
  --This statement executes when every expression in
  --a control statment is false.
  print("Stay indoors.")
end
```

Control Structures

A control structure will run different code blocks based on one or more `true` / `false` statements.

```
if isMonday then
  print("It's the beginning of a great week.")
elseif isWednesday then
  print('Halfway through the week!")
else
  print("Enjoy your day!")
end
```

If statements

A Lua `if` statement is the first part of an if-based control structure. The `if` statement's boolean expression is the first to be evaluated. An `if` statement does not need to be followed by any `else` or `elseif` statements.

```
if weather == "bad" then
  print("Bring a jacket!")
end
```

Elseif Statements

A Lua `elseif` statement allows for additional boolean expressions to be evaluated in an if-based control structure. There is no limit to how many `elseif` statements are in a control structure, but `elseif` statements must be after the `if` statement and before a possible `else` statement.

```
if player == "track star" then
  speed = 10
elseif player == "injured" then
  speed = 1
elseif player == "training" then
  speed = 7
else
  speed = 5
end
```

Code block

A code block is a section of code. Control structures can be set up to determine whether or not they execute.

```
if runCodeBlock then
  -- The following indented code is a
code block:
  print("Part of the code block.")
  print("Also part of the code block.")
end
```

Boolean expressions

Boolean expressions evaluate to either `true` or `false` and are the conditional statements that make up a control structure. You can also include operators and variables in boolean expressions to make your logic more complex.

```
7 > 5 -- Evaluates to: true
```

And operator

The `and` operator is placed between two boolean expressions. It evaluates to `true` when both the left and right sides are `true`. Otherwise the `and` operator evalutes to `false`.

```
true and 5 == 5 -- Evaluates to true
```

or operator

The `or` operator is placed between two boolean expressions and evaluates to `true` if either side is `true`. Otherwise the `or` operator evalutes to `false`.

```
false or 5 == 5  -- Evaluates to: true
```

Not operator

The `not` operator is placed before a boolean expression and evaluates to the inverse of the expression to its right.

```
not ("Hello" == "Hello")  -- Evaluates to: false
```

Comparison operators

Comparison operators compare two variables and either return `true` or `false`. The comparison operators in Lua are as follows: `<`, `>`, `<=`, `>=`, `==`, `~=`.

```
8 >= 10 -- Evaluates to: false
```

Print Share